Is Spanish Really 30% More Expensive?
A Four-Tokenizer Audit of Cross-Lingual Token Arbitrage

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Abstract

This is a benchmark note: one claim, three experiments, verdicts. The claim, from TokenTranslation's design (a middleware that translates prompts into cheaper token spaces): *Spanish costs ∼30% more tokens than English, so translating before the LLM call saves real money.* **E1 (four tokenizers, 15 prompt pairs):** the claim is true and *expiring.* On GPT-4's cl100k_base the saving is 30.2%; on GPT-4o's o200k_base it **halves to 15.8%**; on a multilingual-first vocabulary (XLM-R) it nearly vanishes (7.4%, one prompt already negative); on Qwen2.5 it persists (29.2%). French and Portuguese replicate the halving (33→17%, 31→15%). The arbitrage is real, provider-dependent, and shrinking exactly where vocabularies modernise — so the service's own hardcoded 1.27× multiplier is wrong in both directions at once, and per-request measurement (which it does) is the only durable design. **E2 (backends):** the deployed path delivers 27–35% measured savings at ∼0.6 s warm latency via Google; the *local* Argos translator, never selected in production, proves nearly as accurate on content (12/15 vs. 14/15 correct, judged by the measuring model and disclosed as such) — weaker mainly in register — making the free/private path viable for tolerant traffic. **E3 (tokinensis audit, carried from the previous edition):** the dialect we once called reversible restores **0/30** inputs, corrupts across languages at decode time, and on English input is a net token *cost* 47% of the time. Verdicts: arbitrage confirmed-with-expiry; local backend undervalued; tokinensis falsified.

1 The claim

BPE vocabularies trained on English-heavy corpora fragment other languages into more tokens (Sennrich et al., 2016; Petrov et al., 2023), and commercial pricing bills by the token, so the same meaning costs more in Spanish than in English (Ahia et al., 2023). TokenTranslation operationalises the countermeasure: detect language, translate into the cheapest adequate token space, answer in the user's language, and *measure* the saving per request with a real tokenizer (`tiktoken` cl100k_base as proxy (OpenAI, 2023)). Its code also hardcodes the expectation — a `LANGUAGE_TOKEN_MULTIPLIERS` table asserting Spanish at 1.27× English. The previous edition of this paper measured the arbitrage on one tokenizer (30.2% mean saving, ratio ≈1.43× — already contradicting the hardcoded constant). This note asks the question that decides whether the whole product category has a future: **is the arbitrage a property of language, or of a particular tokenizer generation?**

2 E1 — The same prompts under four tokenizers

Fifteen Spanish/English prompt pairs (the previous edition's battery), re-tokenized under four vocabularies spanning three design eras; French and Portuguese versions of a 5-prompt subset (produced by the measuring model, disclosed) under the two OpenAI vocabularies.

Table 1: Mean saving from translating into English, per tokenizer (n=15 pairs; fr/pt subsets n=5). Same texts throughout.

Tokenizer (era)	**es→en**	**fr→en**	**pt→en**
cl100k_base (GPT-4, 2023)	**30.2%**	33.0%	31.4%
Qwen2.5 (2024)	29.2%	—	—
o200k_base (GPT-4o, 2024)	**15.8%**	17.4%	15.3%
XLM-R (multilingual-first, 2020)	7.4%	—	—

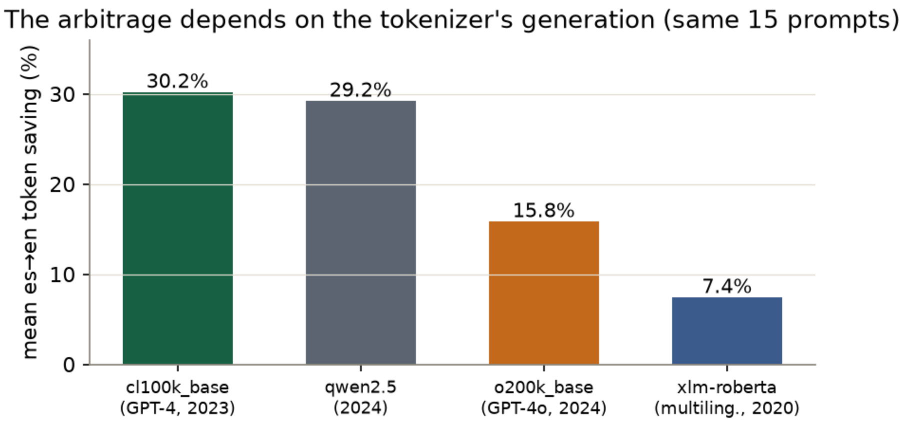

Figure 1: Mean es→en saving across tokenizer generations, same 15 prompts. One provider generation (cl100k→o200k) halves the arbitrage.

The verdict writes itself (Table 1, Figure 1). Moving one generation within the same provider — cl100k to o200k — **halves the arbitrage**, and the halving replicates across all three Romance languages tested. A vocabulary built multilingual-first (XLM-R) leaves almost nothing to arbitrage (7.4% mean, minimum −1.6%: one prompt is already *cheaper* in Spanish). Qwen2.5 shows the effect is not universal — its English-tilted BPE keeps the full margin. Two conclusions follow. For the product: the saving is real today on GPT-4-era and Qwen-family pricing, and evaporating wherever vocabularies modernise — a business built on a closing window, which makes TokenTranslation's per-request measurement its only future-proof component, and its hardcoded 1.27× constant (vs. 1.43× measured on cl100k, 1.19× on o200k) wrong twice over. For the field: published multipliers (Petrov et al., 2023; Ahia et al., 2023) date rapidly; any cost claim needs a tokenizer version stamp.

3 E2 — Does the deployed path deliver, and is the local backend really worse?

Live authenticated calls route to the external Google backend, return correct translations, and report **27.8–35.0%** savings — bracketing E1's offline cl100k mean, corroborating the per-request accounting — at 4.1 s cold / **0.55–0.6 s warm** (n=3; indicative). The router never selected the local Argos translator in production, so we benchmarked it directly against Google on all 15 prompts. Content accuracy: **Argos 12/15 vs. Google 14/15** correct (judged by the measuring model against the references — a disclosed, directional judgment, not BLEU/COMET); Argos's failures were mainly *register* (losing the imperative mood of instructions), not facts. For batch and background traffic that tolerates flat register, the free, offline, private backend is close enough that routing it by default — reserving Google for interactive or nuance-sensitive calls — would remove the external dependency from most requests. The router's local-first philosophy (hibrid project, 2026) is currently config, not behaviour; E2 says the config is leaving value unused.

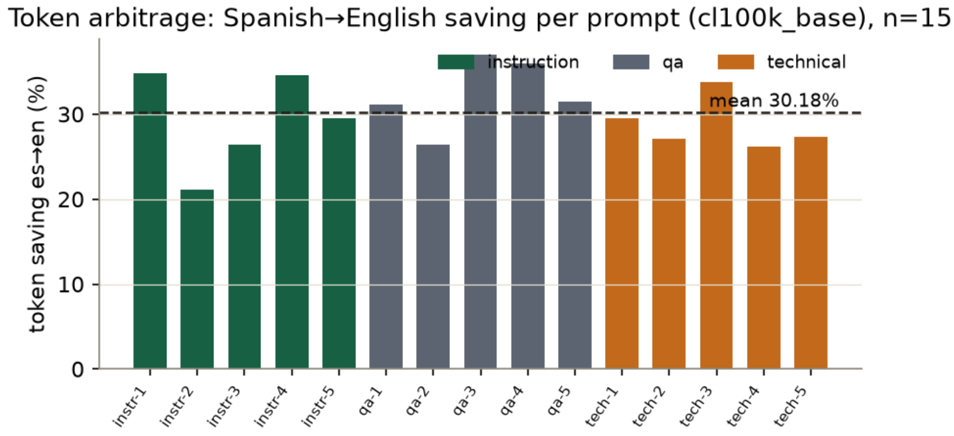

Figure 2: Per-prompt es→en saving on cl100k_base (n=15): every prompt cheaper in English; dashed line = mean 30.2%.

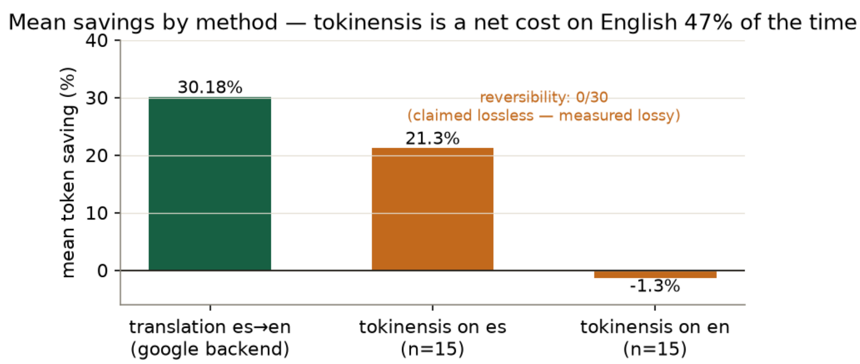

Figure 3: Mean saving by method and input language. Translation: 30.2% meaning-preserving. tokinensis: 21.3% on Spanish, net cost on English, 0/30 reversible.

4 E3 — The tokinensis audit (verdict carried forward)

The service ships *tokinensis*, a deterministic abbreviation dialect once described — by the first edition of this very paper — as reversible. The audit stands: **0/30** exact round-trips. Encoding destroys information by construction (prefix truncation, accent stripping); decoding *adds* damage via unconditional cross-language root substitution (Spanish *"del"* → *"delete"*). Savings: 21.3% on Spanish, **−1.3% on English** (net cost in 7/15 cases) — against the code's hardcoded 0.72× multiplier, wrong like its sibling (Figure 3). Repair paths remain as published: store the inverse or reclassify as lossy; shipping it as reversible is the one option the data forecloses.

5 Verdicts

Claim	Verdict	Evidence
"Spanish costs ∼30% more"	**Confirmed, with expiry**	30.2% (cl100k) → 15.8% (o200k)
Hardcoded 1.27× multiplier	**Falsified twice**	1.43× measured (cl100k), 1.19× (o200k)
Deployed path delivers savings	**Confirmed**	27.8–35.0% live, ∼0.6 s warm
Local backend inadequate	**Unsupported**	Argos 12/15 vs. Google 14/15 content-correct
tokinensis is reversible	**Falsified**	0/30; decode corrupts; −1.3% on English

Limitations. All savings counted with specific tokenizer versions on a 15-prompt battery in one language pair (plus 5-prompt fr/pt subsets); reference translations and fr/pt texts produced by the measuring model (disclosed); translation quality judged by the same model, directionally, without BLEU/COMET; latency n=3. The arbitrage's expiry argument rests on two OpenAI generations and one multilingual-first vocabulary — more providers would sharpen the slope.

Revision note

The first edition asserted properties from design intent; the second measured the arbitrage on one tokenizer and falsified tokinensis' reversibility. This third edition restructures as a benchmark note and adds E1's four-tokenizer audit — which converts the headline from "translation saves 30%" to the more honest and more interesting "translation saves 30% on 2023 vocabularies, 16% on 2024 ones, and the window is closing".

Author note: an AI-made paper

Measured by independent AI agents using the service's own code and public tokenizers; written by AI agents; human owner audited the claims. Published in The PaMaBAI Journal (PaMaBAI editors, 2026).

Artifact

Per-prompt token counts under all four tokenizers, fr/pt subsets, backend comparison transcripts with per-item quality judgments, tokinensis diffs, and external references ship with this paper (`bench/` in `https://github.com/vfalbor/pamabai`).

References

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